Chapter 12: Markov Chain Analysis

1. Introduction

Markov chains model systems that transition between states probabilistically. In learning analytics, states represent engagement levels (Inactive, Passive, Active, Engaged). The transition matrix captures how students move between engagement states, enabling prediction of long-run behavior and identification of critical transitions.

2. Methods

Transition probabilities were estimated from observed state sequences. The steady-state distribution was computed as the left eigenvector of the transition matrix corresponding to eigenvalue 1. N-step transition probabilities were calculated via matrix powers.

3. Results

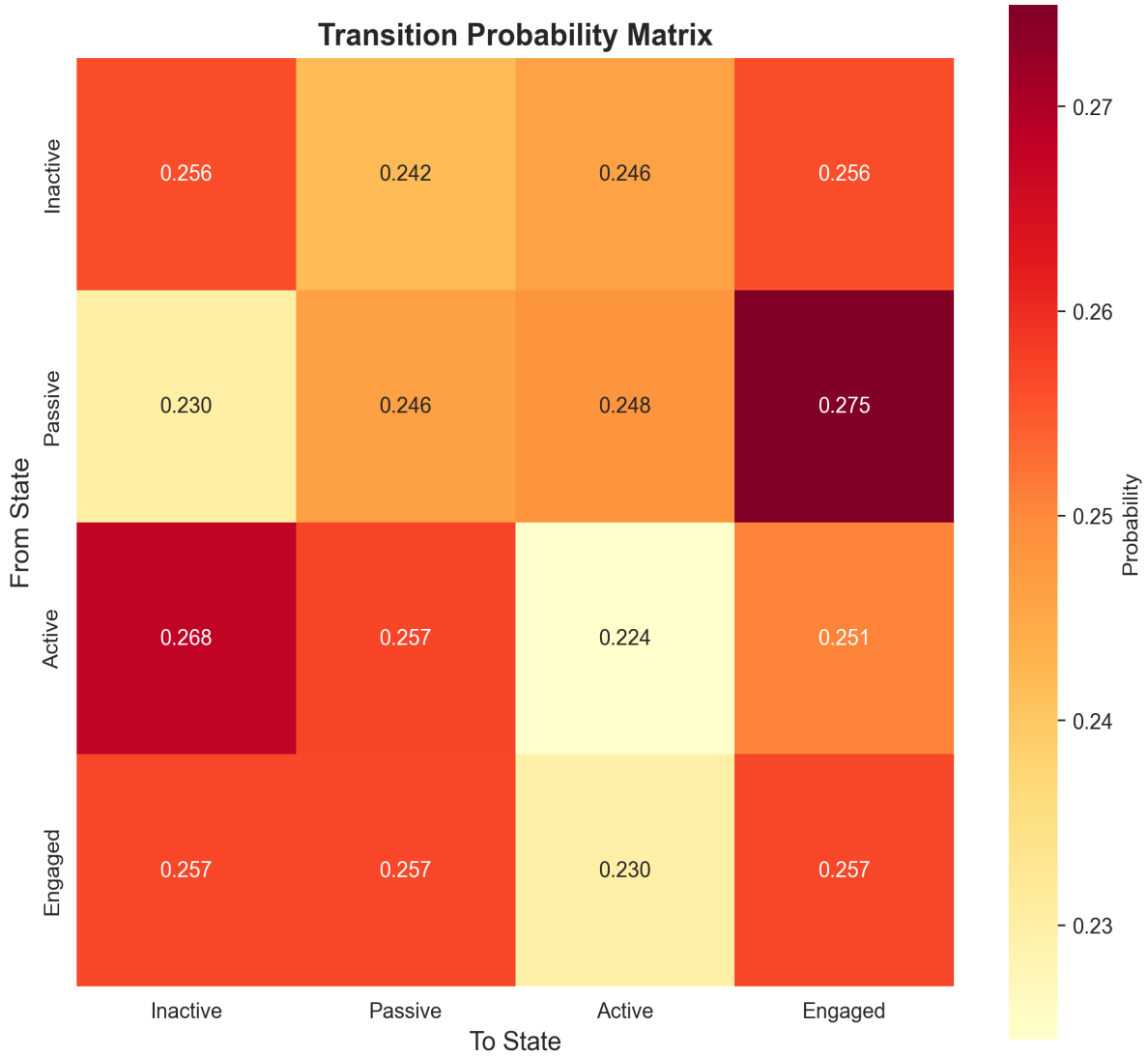

Fig 1: Transition probability matrix.

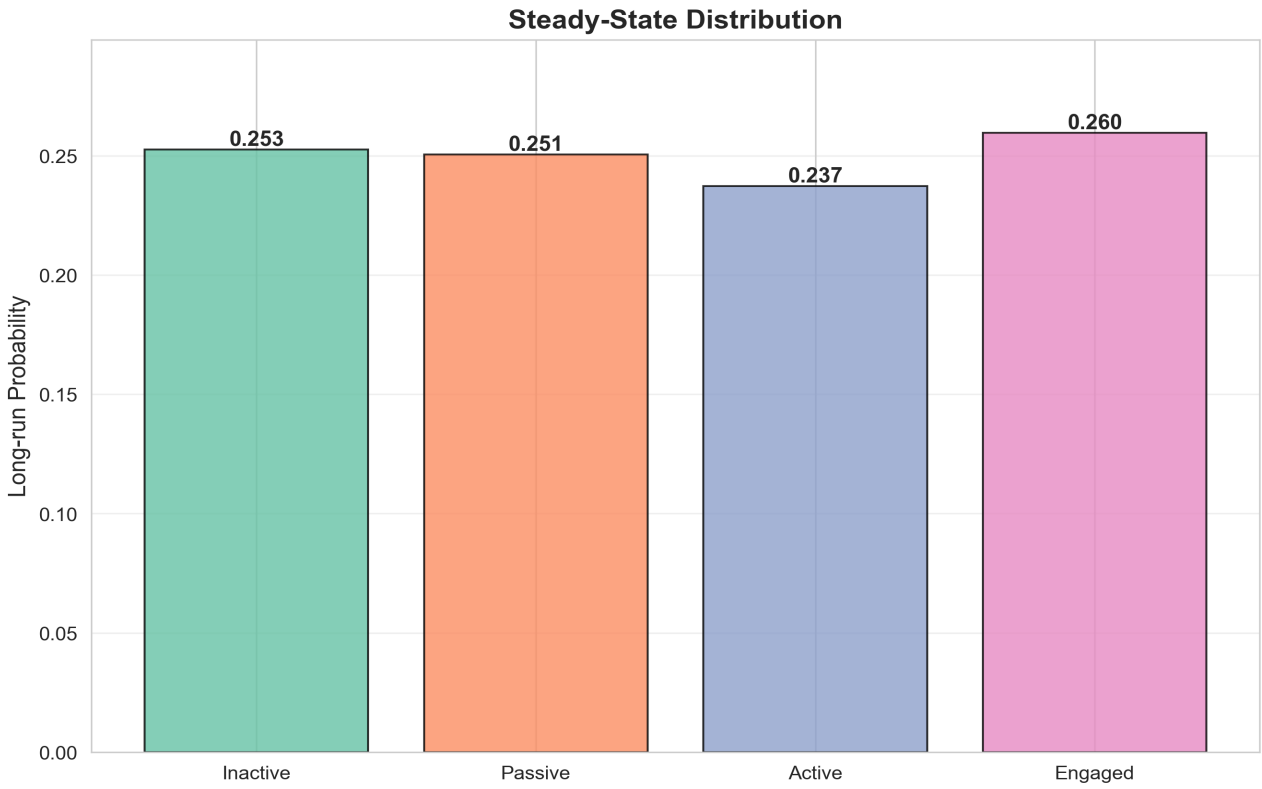

Fig 2: Long-run state distribution.

4. Interpretation

The steady-state distribution shows the expected long-run proportion of time spent in each state. High probabilities of staying "Engaged" indicate course momentum, while high transition probabilities to "Inactive" suggest dropout risk.

5. Limitations

Markov property assumes memorylessness (current state depends only on previous state, not full history). This may not hold for complex learning behaviors. Transition matrix estimated from limited data may be unstable.

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